

Indraprastha Institute of Information Technology, Delhi

End- Semester Examination

EEE Course : ESC 207 A Evolution Ecology and Environment

(02/05/2026)

MM 30

Time 2 hrs.

Attempt all the questions

Fill in the blanks

- I. Dispersal involves the one-way permanent movement of individuals from their birth into another population.
- II. Migrating individuals are classed as emigrants (.....).
- III. is the science of determining the fluxes of materials (e.g. carbon, phosphorus) between different pools (e.g., tree biomass, soil organic material).
- IV. Food eating habit wise hedgehogs are
- V. Heliotropism is observed by the
by moving their heads from east to west to follow the sun's path during the day to maximize sunlight for growth.
- VI. Ethology is the study of observable..... in animals.
- VII. Birds return to their nests—even after migrating thousands of miles—by using a combination of natural navigation tools, including sensing the....., and recognizing visual landmarks.
- VIII. Molecular ecology is closely related to through the study of MABs.
- IX. Ecological engineers restore degraded land by applying scientific and technical knowledge to.....
- X. Seeds contain the package genetic information and nutrients—.....—necessary for the embryo to germinate and grow into a new tree.
- XI. Pine (*Pinus halepensis*) cannot germinate until after their seeds have lived through a
- XII. The most recent replacement for the HCFC is developed by the Honeywell, US company.
- XIII. The Prussian polymath, is honoured in the scientific names of numerous species, including the South American bat (*Phyllostomus hastatus*) and the monkey (*Saimiri sciureus cassiquiarensis*).
- XIV. LULC stands forin EIA .
- XV. EIA stands for

xvi. EC from the MOEFCC, GOI stands for

xvii. Restoration of ecological damage by ecological engineers is termed as or like to like in xviii. A buffer zone in EIA studies is a designated, managed area surrounding a core project site or protected area (often a.....

radius).

xix. In biodiversity gradient there are more species warm and humid climate and decreasing towards drier, colder and more seasonal polar systems or areas. This pattern is often calledDiversity Gradient.

xx. Perch (*Perca fluviatilis*) shift from feeding on to benthic invertebrates to fish as they grow from larva to adult.

xxi. ~~Seeds contain the package genetic information and nutrients.....necessary for the embryo to germinate and grow into a new tree.~~ BSC norms are

xxii. Conditions become hypoxic (an environment with O₂ concentration below 2 mg/litre) and eventually completely anoxic where anaerobic bacteria thrive among the roots when the soil is subjected to waterlogging, prolonged flooding, or severe compactions.

2. Answer True or False.

i. Dispersal is usually distinguished from migration.

ii. Saturn being the closest planet to the Sun, experiences the most intense solar wind in the solar system.

iii. Smartest animal in the forest is Honey Badger/ Among the most intelligent and capable problem-solvers in the animal kingdom.

iv. Holism means over all patterns of a whole functional system, such as an ecosystem, cannot be predicted or understood by a simple summation of the parts.

v. Population principle of growth

$$dN(t)/dt = rN(t) - \alpha(t)^3$$

$$\text{i.e., } \alpha t^2 = rN(t)(K - N(t))/K.$$

vi.. Reaction in Light : $3\text{CO}_2 + 6\text{NADPH} + 5\text{H}_2\text{O} + 9\text{ATP} \rightarrow \text{G3P} + 2\text{H}^+ + 6\text{NADP}^+ + 9\text{ADP} + 8\text{P}_i$

vii. Metaphysical holism is identified in the trend of increased exterior thickness in shells of different species.

viii. Circadian rhythms are physical, mental, and behavioural changes that follow a 24-hour cycle.

ix. The phenomenon, where a parasite loses its own functional genes and instead acquires or relies on host genes/functions, is a well-documented, advanced stage of evolutionary adaptation known as genome sensitisation combined with vertical gene transfer- Antagonism or parasitism. x. Direct mutualisms occur where the organisms live apart. For example,

trees living in the equatorial regions of the planet supply oxygen into the atmosphere that sustains species living in distant polar regions of the planet. During winter when trees at other places shed leaves.

xi. As the population becomes more crowded, it approaches the island's carrying capacity, thus forcing individuals to compete more heavily for fewer available resources. Under crowded conditions, the population experiences density-dependent forces of natural selection, called r-selection.

xii. K-selected species produce offspring that each have a lower probability of survival to maturity. Although not always the case, K-selection is more common in larger animals, like whales or elephants, with longer lifespans.

xiii. A sea sponge in Antarctica has lived for 15,000 years or that some corals in Hawaii have lived for almost 5,000?

xiv. What is. Ontogenetic niche shift (abbreviated ONS) is an ecological phenomenon where an organism (usually an animal) changes its colour and voice during its ontogeny (development).

xv. ~~Under crowded conditions, the population experiences density-dependent forces of natural selection, called r-selection.~~ Salt water plants are known as halophytes

xvi. Parrots are omnivores.

xvii. Environmentally triggered germination of seeds is called serotony.

xviii. IPCC stands for Intergovernmental Protector of Climate Change.

xix. Methane has 34 times more greenhouse effect than carbon dioxide. False

About 55 % of energy is used in cellular respiration.

xx. Temperate deciduous forests have the maximum Net Primary Productivity.

xxi. Riparian zone is the land adjacent to rivers, streams, lakes, or wetlands.

xxii.. Synecology is the study of a single species in an ecosystem.

3. Answer the following within 50 -100 words for each question. (1.5 marks each part)

- i. Biodiversity is the engine of economy. About 55 % of global GDP and about 33 % of Indian GDP is dependant on biodiversity. Describe the 8 advantages of biodiversity.
- ii. What are the ecological and environmental damages of mining?
- iii. What is the role of AI-MI in the control and restoration of environmental and ecological damage?
- iv. What is plastic rain?

4. (a). If the utilities generate electricity using 33 % efficient coal fired power plant. As a carbon reducing measure if electric water heater that converts heat into electricity at 100 % efficiency are replaced with a gas water heaters with a 75 % conversion efficiency. By what fraction the (and percentage) carbon emission would be reduced.

Emission factor for coal is 25.2×10^6 ton. C/ quad.

Emission factor for gas is 14.5×10^6 ton C/quad.

You may Assume 1 quad of electricity is being delivered to electric heaters.

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4(b). Estimate the tons of carbon in the atmosphere corresponding to a concentration of 390 ppm of CO₂. Assume the total mass of air equals 5.1×10^{18} kg. The density of air at the standard temperature and pressure (STP, 0 ° C and 1 atm.) is 1.29 kg/m^3 .

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